POST-ROUTE PVT CORNER TIMING ANALYSIS REPORT

VSDBabySoC Project | Roll No: 24EG104C36

1. Objective

In this experiment I checked the timing of the VSDBabySoC design after routing. The check was done at 16 different Process, Voltage and Temperature (PVT) corners using the timing engine inside OpenROAD.

2. Overview

Static Timing Analysis (STA) checks that every signal in the circuit reaches its destination on time. My earlier STA was done on the synthesized netlist. This time I used the routed design (**5_route.odb**) and its SDC file (**5_route.sdc**). The wire parasitics were taken from global routing using **estimate_parasitics -global_routing**, so wire delay is now counted.

There are 16 corners: 6 FF, 8 SS and 2 TT. For each corner I noted the worst setup slack, worst hold slack, WNS, TNS and whether setup was met. The PLL and DAC libraries (avsdpll and avsddac) were loaded for every corner.

3. PVT Corner Comparison Table

S.No	PVT Corner	Min Slack (ns)	Max Slack (ns)	WNS (ns)	TNS (ns)	Status
1	FF_100C_1v65	0.268	8.011	0.000	0.000	MET
2	FF_100C_1v95	0.212	8.924	0.000	0.000	MET
3	FF_n40C_1v56	0.315	6.875	0.000	0.000	MET
4	FF_n40C_1v65	0.277	7.587	0.000	0.000	MET
5	FF_n40C_1v76	0.245	8.211	0.000	0.000	MET
6	FF_n40C_1v95	0.205	8.929	0.000	0.000	MET
7	SS_100C_1v40	0.929	-3.514	-3.510	-326.350	**VIOLATED**
8	SS_100C_1v60	0.658	1.615	0.000	0.000	MET
9	SS_n40C_1v28	1.850	-29.322	-29.320	-14575.390	**VIOLATED**
10	SS_n40C_1v35	1.373	-16.453	-16.450	-6570.440	**VIOLATED**
11	SS_n40C_1v40	1.149	-10.985	-10.990	-3336.450	**VIOLATED**
12	SS_n40C_1v44	1.008	-7.720	-7.720	-1909.840	**VIOLATED**
13	SS_n40C_1v60	0.679	-0.300	-0.300	-1.190	**VIOLATED**
14	SS_n40C_1v76	0.505	3.228	0.000	0.000	MET
15	TT_025C_1v80	0.330	6.831	0.000	0.000	MET
16	TT_100C_1v80	0.330	6.957	0.000	0.000	MET

Min Slack is the worst hold slack and Max Slack is the worst setup slack. Status is for setup. Hold is met at all 16 corners.

4. Timing Graphs

These graphs show WNS, TNS, setup slack and hold slack for the 16 corners. The value is written at each point. The corner numbers are the same as the S.No in the table above.

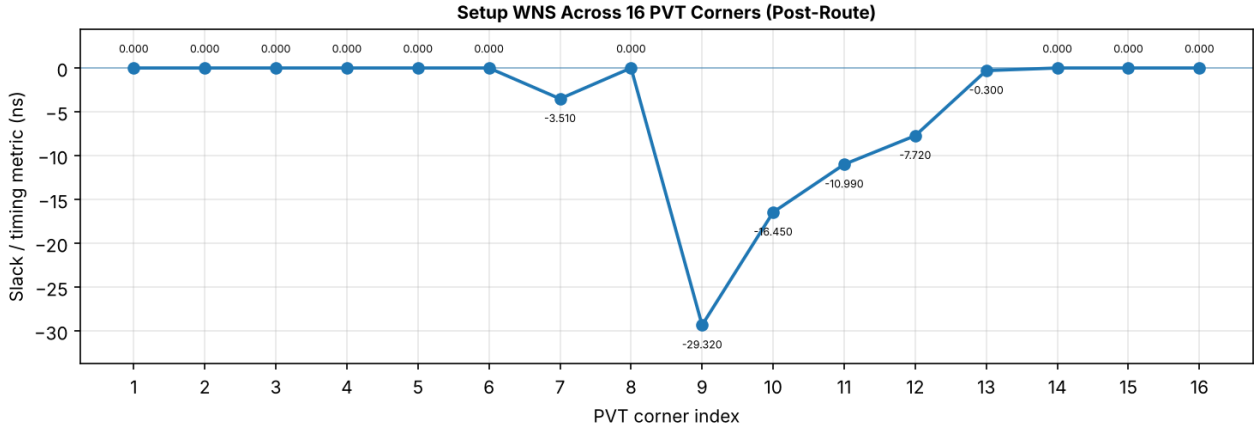


Figure 1: Worst Negative Slack (WNS) across PVT corners

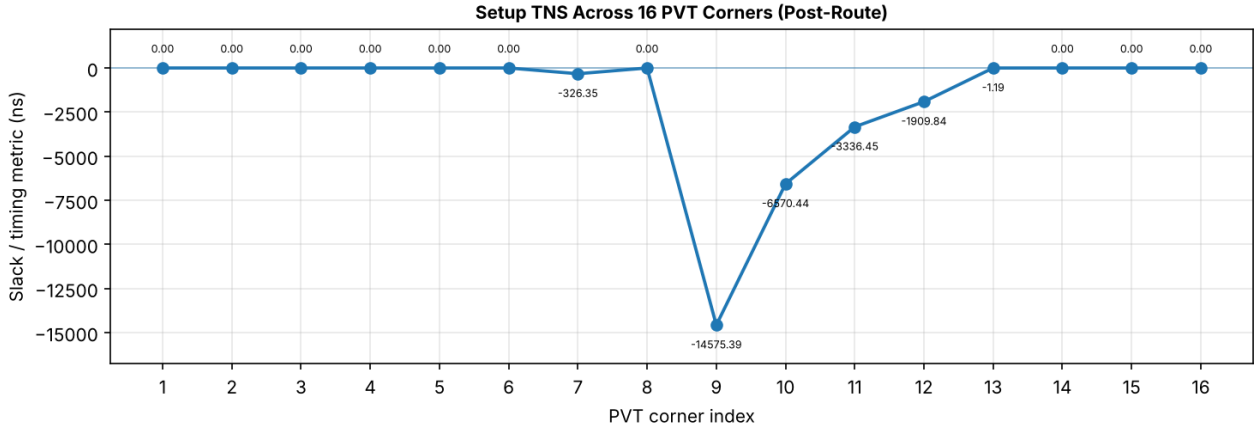


Figure 2: Total Negative Slack (TNS) across PVT corners

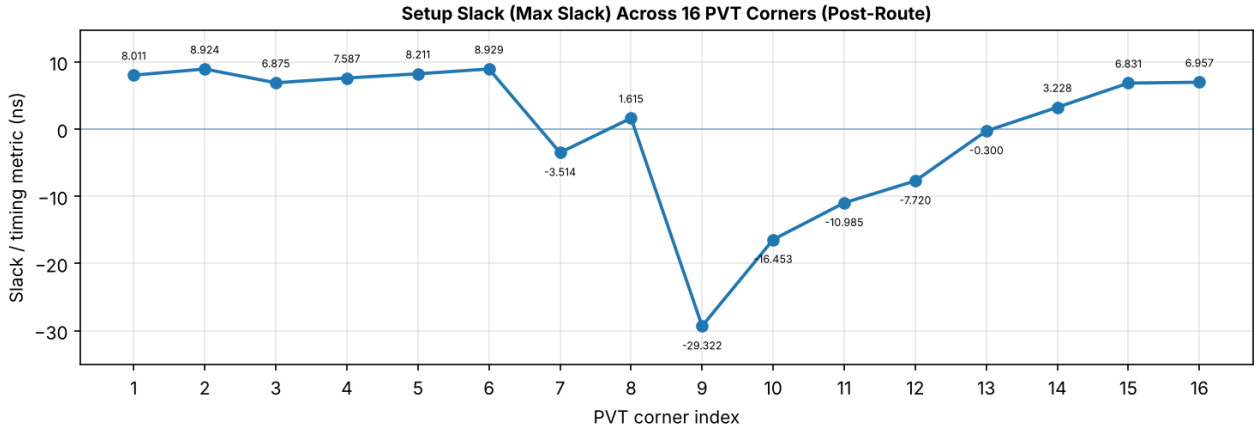


Figure 3: Setup Slack across PVT corners

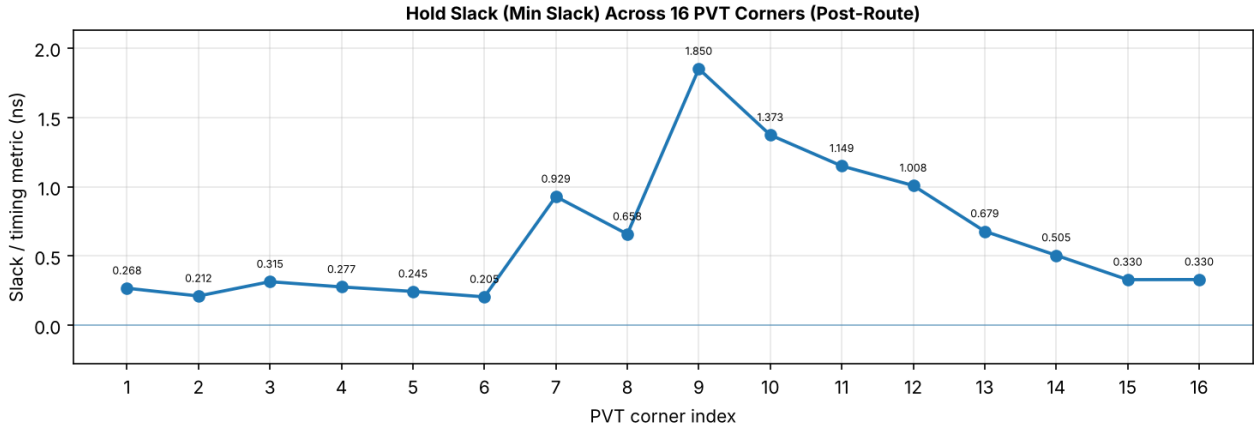


Figure 4: Hold Slack across PVT corners

5. Observations

1. All 6 FF corners pass setup and hold. Their setup slack is between 6.88 and 8.93 ns.
2. Both TT corners also pass. Setup slack is 6.83 ns and 6.96 ns.
3. 6 of the 8 SS corners fail setup. Only SS_100C_1v60 (+1.615 ns) and SS_n40C_1v76 (+3.228 ns) pass.
4. The worst corner is SS_n40C_1v28. Its WNS is -29.322 ns and TNS is -14575.39 ns. SS_n40C_1v60 fails by a small amount, only 0.300 ns (TNS -1.19 ns).
5. For the SS_n40C corners, setup slack gets worse as the supply voltage goes down: +3.23 ns at 1.76 V, -0.30 ns at 1.60 V, -7.72 ns at 1.44 V, -10.99 ns at 1.40 V, -16.45 ns at 1.35 V and -29.32 ns at 1.28 V.
6. At the same 1.40 V, the -40C corner (-10.99 ns) is worse than the 100C corner (-3.51 ns). So at low voltage, the cold corner is slower than the hot one.
7. Hold is met at every corner. The smallest hold slack is +0.205 ns at FF_n40C_1v95. The slow, low-voltage corners have the most hold slack (up to +1.850 ns).

6. Conclusion

I ran post-route STA on the VSDBabySoC design at 16 PVT corners. Setup and hold are met at all FF and TT corners. They are also met at SS_100C_1v60 and SS_n40C_1v76. The remaining 6 SS corners fail setup. These run at 1.28 V to 1.60 V (-40C) and 1.40 V (100C), which is much lower than the 1.80 V normal supply. There is no hold violation at any corner.

To fix the failing corners, the clock period in the SDC can be increased, the long adder and carry path can be made shorter, or bigger cells and buffers can be used on that path.

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